

Weak-instrument Robust Inference with Heterogeneous Treatment Effects*

Arnstein Vestre

University of Chicago

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Abstract

TBC – Rewrite.

*Email: arnsteinvestre@gmail.com.

1. INTRODUCTION

Instrumental-variables (IV) methods are a workhorse of empirical economics. Originally developed in the simultaneous-equations tradition of [Anderson and Rubin \(1949\)](#) and refined for macroeconomic identification, the IV toolkit has, over the last three decades, migrated to applied microeconomics, where it underwrites a large empirical literature on labor markets, education, judges, examiners, and health ([Angrist and Imbens, 1995](#); [Angrist et al., 2000](#); [Baioocchi et al., 2014](#)). Many of those applications appeal to multiple instruments to gain power: judge fixed effects, lottery-induced variation, shift-share decompositions, and Mendelian-randomization sets routinely involve dozens or hundreds of instruments. At the same time, applied work has moved away from the constant-effects model that motivated the classical IV asymptotics. Causal effects are presumed heterogeneous, and the two-stage least-squares (TSLS) estimand is interpreted as a positively-weighted average of local average treatment effects (LATEs), as in [Angrist and Imbens \(1995\)](#), [Kolesár \(2013\)](#), and [Mogstad et al. \(2021\)](#).

This combination — many instruments, weak first stages, and heterogeneous treatment effects — is now the empirical default rather than the exception. Yet the workhorse weak-instrument-robust tests, against which applied researchers benchmark TSLS confidence intervals, were not designed for it. The [Anderson and Rubin \(1949\)](#) AR test and its descendant, the [Moreira \(2003\)](#) CLR test, retain their nominal size under weak instruments only when treatment effects are constant. As soon as the data-generating process admits heterogeneous responses to the instruments — the very feature that justifies the LATE interpretation of TSLS — these tests over-reject, and the size distortion grows with instrument strength. The result is a gap in the inference toolkit precisely where applied work most often sits.

Why the AR/CLR tests fail. The AR test inverts the moment restriction $\mathbb{E}[Z_i(Y_i - \beta_0 D_i)] = 0$, which is equivalent to the joint hypothesis that (a) there exists a scalar β such that the reduced form satisfies $\delta_n = \beta \gamma_n$, and (b) that β equals the hypothesized β_0 . With more than one instrument, (a) is a substantive restriction: it is the constant-effects assumption in disguise. Treatment-effect heterogeneity violates (a), so the AR test rejects even when (b) is true. The decomposition $\text{AR}(\beta_0) = \text{LM}(\beta_0) + J(\beta_0)$ ([Kleibergen, 2007](#)) makes this transparent: the J component is a Sargan–Hansen test of overidentification ([Sargan, 1958](#); [Hansen, 1982](#)), which under strong exogeneity rejects whenever $\delta_n - \beta \gamma_n \neq 0$.

The CLR test (Moreira, 2003) inherits the AR statistic through its construction and so inherits this failure mode. The LM test is robust to (a) but loses size under finite concentration parameters when heterogeneity is present.

This paper. We propose a weak-instrument-robust test for the TSLS estimand $\beta = \gamma'_n \Sigma_{ZZ} \delta_n / \gamma'_n \Sigma_{ZZ} \gamma_n$ that retains its size under treatment-effect heterogeneity, weak instruments, and arbitrary numbers of instruments. The test is the asymptotic likelihood ratio for $H_0 : \beta = \beta_0$, treating H_0 as a quadratic constraint on the reduced-form/first-stage coefficient pair $\tau_n \equiv (\delta'_n, \gamma'_n)'$. The constrained quasi-likelihood is a generalized trust-region subproblem in the sense of Stern and Wolkowicz (1995) and admits a closed-form solution at the optimum.

The resulting statistic, which we call TLR (TSLS likelihood ratio), has a tractable limiting distribution under strong exogeneity. The distribution depends on two scalar nuisance parameters: a magnitude-share $\varrho \in (0, 1)$ governing the relative size of the positive and negative eigenvalues of the constraint matrix, and a non-centrality $\xi \geq 0$ summarizing instrument strength, treatment-effect heterogeneity, and endogeneity. We show that ϱ has a consistent estimator and that ξ admits a sufficient statistic with a chi-squared limit. With these in hand we construct two valid tests: a least-favorable-configuration (LFC) test that takes the worst-case quantile over ξ , and a Berger–Boos two-step test (Berger and Boos, 1994) that first builds a confidence interval for ξ and then takes the worst-case quantile over that interval. The Berger–Boos variant retains nontrivial power under both weak and strong instruments and, in our simulations, has size close to nominal and power close to the (size-distorted) TSLS Wald test that researchers currently use without correction.

The contribution is methodological: under standard heterogeneous-effects identification (Angrist and Imbens, 1995; Angrist et al., 2000) and the weak-instrument asymptotic of Staiger and Stock (1997), our test is, to our knowledge, the first that delivers correct size for the TSLS estimand without requiring constant effects, a single instrument, or many-instrument asymptotics. The Section 5 extension drops the strong-exogeneity restriction in favor of mean exogeneity and characterizes the limiting distribution in that setting; the construction continues to apply but with a larger nuisance space.

Related literature. Our paper sits at the intersection of three literatures: (i) weak-instrument-robust inference under fixed instrument count, (ii) inference with many weak

instruments, and (iii) identification under heterogeneous treatment effects.

The fixed- d_Z weak-instrument literature begins with [Anderson and Rubin \(1949\)](#) and was revived by [Staiger and Stock \(1997\)](#), [Bound et al. \(1995\)](#), and the testing program of [Stock et al. \(2002\)](#) and [Stock and Yogo \(2005\)](#). The K-test of [Kleibergen \(2007\)](#), the CLR test of [Moreira \(2003\)](#), the optimal invariant tests of [Andrews et al. \(2006\)](#), and the practical implementations of [Finlay and Magnusson \(2009\)](#) all maintain constant treatment effects. [Moreira \(2009\)](#) and [Andrews \(2016\)](#) extend the conditional approach but, again, under constant effects. We share the conditional-inference perspective and the use of likelihood-ratio statistics, but we drop the constant-effects assumption and so live in a different limiting-distribution family — one indexed by (ϱ, ξ) rather than by the single concentration parameter μ^2 .

The many-weak-instrument literature ([Chao and Swanson, 2005](#); [Chao et al., 2012](#); [Mikusheva and Sun, 2022, 2024](#); [Yap, 2025](#)) delivers tests that are valid under heterogeneous effects but only when $d_Z \rightarrow \infty$ along with n . The jackknife LM test of [Yap \(2025\)](#) is the closest existing procedure to ours in spirit, though its asymptotics are taken in a different regime. Heteroskedasticity-robust inference with fixed d_Z traces to [Kolesár \(2013\)](#) and [Evdokimov and Kolesár \(2018\)](#); we maintain a heteroskedasticity-consistent variance estimator throughout.

The heterogeneous-effects identification literature begins with [Angrist and Imbens \(1995\)](#); the discrete-instrument LATE framework of ? and the IV-as-LATE-aggregator argument of [Angrist et al. \(2000\)](#) and [Mogstad et al. \(2021\)](#) make explicit the conditions under which TSLS targets a meaningful causal estimand. For the judge/examiner application that motivates much of our empirical interest, see ? and ?, who document the prevalence of the multi-instrument, heterogeneous-effects regime in applied work.

Roadmap. Section 2 sets up the model and notation. Section 3 reviews the AR, LM, and CLR tests and shows how each loses size under heterogeneity. Section 4 introduces the TLR statistic, derives its limiting distribution, and constructs both the LFC and Berger–Boos tests. Section 5 drops strong exogeneity. Section 6 concludes. Proofs are collected in Appendix A.

2. SET-UP AND NOTATION

Fundamentals. Let $\{(Y_i, D_i, Z_i)\}_{i=1}^n$ be an i.i.d. sample, where Y_i is an outcome, D_i a treatment, both scalar, and Z_i a vector of d_Z instruments. Assume, without loss of generality, all variables are demeaned and have finite fourth moment. In the presence of covariates, let Y_i and D_i denote variables where the impact of the covariates is appropriately projected off. We are interested in the linear heterogeneous treatment effects model,

$$Y_i = \beta_i D_i + U_i \quad (1)$$

where β_i is the causal effect of D_i on Y_i , and U_i is the model residual. Denote the linear regressions of Y_i on Z_i (reduced form) and D_i on Z_i (first stage) as,

$$Y_i = \delta'_n Z_i + W_i \quad (2)$$

$$D_i = \gamma'_n Z_i + V_i \quad (3)$$

where γ_n is the vector of first-stage coefficients, δ_n the vector of reduced-form coefficients, and V_i and W_i are regression residuals.

The subscripts on γ_n and δ_n indicate that we do not restrict the data generating process to be invariant with respect to the sample size n . In particular, in order to analyze weak-instrument properties of our procedures, we follow [Staiger and Stock \(1997\)](#) in letting (δ_n, γ_n) drift towards zero at some rate $(\bar{\delta}, \bar{\gamma})$, such that $\delta_n \equiv \bar{\delta}/\sqrt{n}$ and $\gamma_n \equiv \bar{\gamma}/\sqrt{n}$. In order to study the properties of our test with strong instruments, we hold (δ_n, γ_n) fixed as $n \rightarrow \infty$. We maintain the following assumptions.

Assumption IVR (Relevance). $\mathbb{E}[Z_i D_i] \neq \mathbf{0}_{d_Z}$

Assumption IVX (Strong Exogeneity).

- (a) If D_i is continuous, assume $(\beta_i, U_i, V_i) \perp\!\!\!\perp Z_i$
- (b) If D_i is binary, assume $(\beta_i, U_i, \{D_i(z)\}_{z \in \mathbb{R}^{d_Z}}) \perp\!\!\!\perp Z_i$

where $D_i(z)$ denotes the treatment of an individual with instrument realization $Z_i = z$.

Observe that [Assumption IVX](#) is stronger than the (weak) exogeneity usually assumed in a constant effects model, $\mathbb{E}[Z_i U_i] = \mathbf{0}_{d_Z}$. However, it is equivalent to the assumptions usually maintained in order to obtain identification of models with heterogeneous treatment effects, either in the binary (see e.g. [Angrist and Imbens, 1995](#)) or continuous treatment case (see e.g. [Angrist et al., 2000](#)). However, no assumption is imposed on the monotonicity (uniformity) of $D_i(z)$ in z .

Observed moments. Let $(\hat{\delta}, \hat{\gamma})$ be OLS estimators of (δ_n, γ_n) . The two estimators have the following joint limiting distribution.¹

$$\begin{bmatrix} \sqrt{n}(\hat{\delta} - \delta_n) \\ \sqrt{n}(\hat{\gamma} - \gamma_n) \end{bmatrix} \xrightarrow{d} \mathfrak{N}(\mathbf{0}_{2d_Z}, \Sigma) \quad \Sigma \equiv \begin{bmatrix} \Sigma_{\hat{\delta}} & \Sigma_{\hat{\delta}, \hat{\gamma}} \\ \Sigma'_{\hat{\delta}, \hat{\gamma}} & \Sigma_{\hat{\gamma}} \end{bmatrix} \quad (4)$$

It will later be convenient to write $\tau_n \equiv \begin{bmatrix} \delta_n \\ \gamma_n \end{bmatrix}$ and $\hat{\tau} \equiv \begin{bmatrix} \hat{\delta} \\ \hat{\gamma} \end{bmatrix}$, and the limiting distribution as $\sqrt{n}(\hat{\tau} - \tau_n) \xrightarrow{d} \mathfrak{N}(\mathbf{0}_{2d_Z}, \Sigma)$.

The TSLS Hypothesis. Let β denote the two-stage least squares (TSLS) estimand, a positively weighted average of local average treatment effects (LATEs), targeted by the TSLS and Jackknife IV estimators (Kolesár, 2013). It is defined as follows.

$$\beta \equiv \frac{\gamma'_n \Sigma_{ZZ} \delta_n}{\gamma'_n \Sigma_{ZZ} \gamma_n} \quad (5)$$

where $\Sigma_{ZZ} \equiv \mathbb{E}[Z_i Z'_i]$ denotes the population gram matrix of Z_i . The TSLS estimand admits a causal interpretation under certain assumptions, see e.g. Angrist and Imbens (1995) and Mogstad et al. (2021). Hypotheses about the estimand, $\beta = \beta_0$, can be rewritten as a quadratic constraint, $\gamma'_n \Sigma_{ZZ} (\delta_n - \beta_0 \gamma_n) = 0$. In terms of τ_n , we obtain,

$$\tau'_n \hat{\Gamma}(\beta_0) \tau_n = 0 \quad \text{where} \quad \Gamma(\beta_0) \equiv \begin{bmatrix} 0 & \Sigma_{ZZ} \\ \Sigma_{ZZ} & -2\beta_0 \Sigma_{ZZ} \end{bmatrix} \quad (6)$$

Eigenvalue Representation. Define the rotation of $\Gamma(\beta_0)$ in terms of Σ as follows.

$$\Lambda(\beta_0) \equiv \Sigma^{\frac{1}{2}} \Gamma(\beta_0) \Sigma^{\frac{1}{2}} \quad (7)$$

$\Lambda(\beta_0)$ is an indefinite matrix. However, the block-structure of $\Gamma(\beta_0)$ imposes structure on the sign of its eigenvalues. Under [Assumption IVX](#), the structure simplifies further.

Lemma EH (Eigenvalue Homogeneity). $\Lambda(\beta_0)$ has d_Z positive and d_Z negative eigenvalues. Let $\{\kappa_-^{(k)}, \kappa_+^{(k)}\}_{k=1}^{d_Z}$ denote these eigenvalues.

1. Under [Assumption IVX\(a\)](#), $\kappa_-^{(k)} = \kappa_-$ and $\kappa_+^{(k)} = \kappa_+$ for all $k = 1, \dots, d_Z$
2. Under [Assumption IVX\(b\)](#), $\kappa_-^{(k)} \rightarrow \kappa_-$ and $\kappa_+^{(k)} \rightarrow \kappa_+$ as $n \rightarrow \infty$ and $\gamma_n \sqrt{n} = \bar{\gamma}$ for all $k = 1, \dots, d_Z$

¹Throughout, $\mathfrak{N}(\mu, \Sigma)$ denotes a (multivariate) normal distribution with mean μ and variance Σ , and $\mathfrak{C}_k^2(\lambda)$ denotes a non-central chi-squared distribution with k degrees of freedom and non-centrality parameter λ (central when λ is omitted).

Let $\varrho \equiv \kappa_+ / (\kappa_+ + |\kappa_-|)$ denote the magnitude share of the positive eigenvalue.

Proof. See Appendix A.1. ■

Let X denote the matrix of eigenvectors of $\Lambda(\beta_0)$, Define the rotated first-stage and reduced-form estimators and estimands as follows.²

$$\hat{W} \equiv \hat{X}' \hat{\Sigma}^{-1/2} \sqrt{n} \hat{\tau} \quad \text{and} \quad w \equiv X' \Sigma^{-1/2} \sqrt{n} \tau_n \quad (8)$$

The vector of statistics $\hat{W}$ has the following limiting distribution under the null.

$$\hat{W} - w \xrightarrow{d} \mathfrak{N}(\mathbf{0}_{2d_Z}, \boldsymbol{\nu}_{2d_Z}) \quad (9)$$

and we can write the TSLS hypothesis as $\kappa_+ \|w_+\|^2 = \kappa_- \|w_-\|^2$, where $w_\pm$ denotes the subvector of w associated with positive and negative eigenvalues, respectively.

Sample statistics. To quantify the identifying power of the instrument set, it will be useful to leverage a summary statistic whose limiting distribution is invariant under β_0 . We record the statistic and its limiting distribution as a lemma.

Lemma STAT (Summary Statistic). *Let $\hat{S} \equiv \|(\hat{\Sigma}/n)^{-\frac{1}{2}} \hat{\tau}\|^2 / d_Z$. Then,*

$$d_Z \hat{S} \xrightarrow{d} \mathfrak{C}_{2d_Z}^2(\xi)$$

where $\xi \equiv \|(\Sigma/n)^{-\frac{1}{2}} \tau_n\|^2$. Let $\Sigma_{\delta|\hat{\gamma}} \equiv \Sigma_{\delta} - \Sigma_{\delta, \hat{\gamma}} \Sigma_{\hat{\gamma}}^{-1} \Sigma'_{\delta, \hat{\gamma}}$, and let μ^2, ν^2, η^2 denote three scalars, where $\mu^2 \equiv \|(\Sigma_{\hat{\gamma}}/n)^{-\frac{1}{2}} \gamma_n\|^2$ is the concentration parameter, $\nu^2 \equiv \|(\Sigma_{\delta|\hat{\gamma}}/n)^{-\frac{1}{2}} (\delta_n - \beta \gamma_n)\|^2$ captures treatment effect heterogeneity and $\eta^2 \equiv \|(\Sigma_{\delta|\hat{\gamma}}/n)^{-\frac{1}{2}} \gamma_n (\beta_{ols} - \beta)\|^2$ captures, among other things, endogeneity. We then have $\xi = \mu^2 + \nu^2 + \eta^2$.

Proof. See Appendix A.2. ■

Remark 1. Observe that by X and $\hat{X}$ orthogonal, we have $\hat{S} \equiv \|\hat{W}\|^2 / d_Z$ and $\xi \equiv \|w\|^2$. Observe further that μ^2, ν^2 and η^2 are constant along the weak instrument limiting sequence, and can be associated with three statistics,

$$\begin{aligned} \hat{F} &\equiv d_Z^{-1} \|(\hat{\Sigma}_{\hat{\gamma}}/n)^{-\frac{1}{2}} \hat{\gamma}\|^2 & d_Z \hat{F} &\xrightarrow{d} \mathfrak{C}_{d_Z}^2(\mu^2) \\ \hat{J} &\equiv d_Z^{-1} \|(\hat{\Sigma}_{\delta|\hat{\gamma}}/n)^{-\frac{1}{2}} (\hat{B}\hat{\gamma} - \hat{\delta})\|^2 & d_Z \hat{J} &\xrightarrow{d} \mathfrak{C}_{d_Z-1}^2(\nu^2) \\ \hat{H} &\equiv d_Z^{-1} \|(\hat{\Sigma}_{\delta|\hat{\gamma}}/n)^{-\frac{1}{2}} (\hat{\Sigma}_{\delta, \hat{\gamma}} \hat{\Sigma}_{\hat{\gamma}}^{-1} \hat{\gamma} - \hat{B}\hat{\gamma})\|^2 & d_Z \hat{H} &\xrightarrow{d} \mathfrak{C}_1^2(\eta^2) \end{aligned}$$

where $\hat{F}$ is the first stage F -statistic, often used as a diagnostic for instrument strength. ■

²Note: Change from W, X to something else.

3. ON THE VALIDITY OF EXISTING WEAK-INSTRUMENT ROBUST TESTS

In the following, we proceed to show that the [Anderson and Rubin \(1949\)](#) AR test, the [Kleibergen \(2007\)](#) LM test and the [Moreira \(2003\)](#) CLR test do not retain size under the combination of weak instruments and treatment effect heterogeneity, when $d_Z > 1$. We record the definitions of the statistics, in the notation [Finlay and Magnusson \(2009\)](#).

Definition LEG (Legacy Robust Test Statistics). *Let $\hat{\Omega}(\beta_0) \equiv \hat{\Sigma}_{\hat{\delta}} - 2\beta_0 \hat{\Sigma}_{\hat{\delta}, \hat{\gamma}} + \beta_0^2 \hat{\Sigma}_{\hat{\gamma}}$ denote the variance of the restriction $\hat{\delta} - \beta_0 \hat{\gamma}$, and let*

$$\tilde{\gamma}(\beta_0) \equiv \hat{\gamma} - \hat{\Omega}(\beta_0)^{-1} (\hat{\Sigma}_{\hat{\delta}, \hat{\gamma}} - \beta_0 \hat{\Sigma}_{\hat{\gamma}}) (\hat{\delta} - \beta_0 \hat{\gamma})$$

denote the first stage orthogonalized relative to $\hat{\delta} - \beta_0 \hat{\gamma}$.

1. *The AR test statistic is defined as*

$$AR(\beta_0) \equiv \left\| (\hat{\Omega}(\beta_0)/n)^{-1/2} (\hat{\delta} - \beta_0 \hat{\gamma}) \right\|^2 \quad (10)$$

2. *The LM test statistic is defined as*

$$LM(\beta_0) \equiv \left[\tilde{\gamma}(\beta_0)' (\hat{\Sigma}(\beta_0)/n)^{-1} (\hat{\delta} - \beta_0 \hat{\gamma}) \right]^2 / \left\| (\hat{\Sigma}(\beta_0)/n)^{-1/2} \tilde{\gamma}(\beta_0) \right\|^2. \quad (11)$$

3. *The CLR test statistic is defined as*

$$CLR(\beta_0) = \frac{1}{2} \left[AR(\beta_0) + \sqrt{(AR(\beta_0) - \hat{r}(\beta_0))^2 + 4 \cdot LM(\beta_0) \cdot \hat{r}(\beta_0)} \right] \quad (12)$$

where $\hat{r}(\beta_0) \equiv \left\| (\hat{\Sigma}(\beta_0)/n)^{-1/2} \tilde{\gamma}(\beta_0) \right\|^2$. The Sargan-Hansen J -statistic is defined as $\hat{J}(\beta_0) \equiv AR(\beta_0) - LM(\beta_0)$.

With constant effects, we have the following limiting distributions.

Lemma LEG-CTE (Legacy tests, Constant Treatment Effects). *Assume there exists a scalar β such that $\delta_n = \beta \gamma_n$. We then have, uniformly, the following limits.*

1. $AR(\beta_0) \xrightarrow{d} \mathfrak{C}_{d_Z}^2$
2. $LM(\beta_0) \xrightarrow{d} \mathfrak{C}_1^2$
3. $LR(\beta_0) \xrightarrow{d} \frac{1}{2} \left[\mathfrak{C}_{d_Z-1}^2 + \mathfrak{C}_1^2 + \sqrt{(\mathfrak{C}_{d_Z}^2 + \mathfrak{C}_1^2)^2 + 4 \cdot \mathfrak{C}_1^2 \cdot \hat{r}} \right]$

Proof. See Appendix [A.3](#). ■

With treatment effect heterogeneity, this result no longer obtains.

Lemma LEG-HTE (Legacy tests, Heterogeneous Treatment Effects). *Assume there exists a scalar β such that $\delta_n = \beta\gamma_n$. Then, we have:*

1. $\text{AR}(\beta_0) \xrightarrow{d} \mathfrak{C}_{dZ}^2(\lambda_{\text{AR}})$ where $\lambda_{\text{AR}} = \frac{\mu^2}{\mu^2 + \eta^2} \cdot \nu^2$
2. $\text{LM}(\beta_0) \xrightarrow{d} \mathfrak{C}_1^2(\lambda_{\text{LM}})$ where $\lambda_{\text{LM}} = \frac{\eta^2 \mu^2 \nu^2}{(\mu^2 + \eta^2)[(\mu^2 + \eta^2)^2 + \eta^2 \nu^2]} \cdot \nu^2$
3. $\text{J}(\beta_0) \xrightarrow{d} \mathfrak{C}_{dZ-1}^2(\lambda_{\text{J}})$ where $\lambda_{\text{J}} = \frac{\mu^2(\mu^2 + \eta^2)}{(\mu^2 + \eta^2)^2 + \eta^2 \nu^2} \cdot \nu^2$
4. $\text{LR}(\beta_0) \xrightarrow{d} \frac{1}{2} \left[\mathfrak{C}_{dZ-1}^2(\lambda_{\text{J}}) + \mathfrak{C}_1^2(\lambda_{\text{LM}}) + \sqrt{(\mathfrak{C}_{dZ-1}^2(\lambda_{\text{J}}) + \mathfrak{C}_1^2(\lambda_{\text{LM}}))^2 + 4 \cdot \mathfrak{C}_1^2(\lambda_{\text{LM}}) \cdot \hat{r}} \right]$

where μ^2, ν^2 and η^2 are as defined in the previous section.

Proof. See Appendix A.4. ■

Remark 2. *We make two observations.*

1. *AR and CLR are centered away from zero whenever $\nu^2 > 0$, i.e. in the presence of heterogeneous treatment effects. The result holds in the limit as $\mu^2 \rightarrow \infty$.*
2. *LM is centered away from zero whenever $\nu^2 > 0$ and $\mu^2 < \infty$. The test is centered at zero in the limit as $\nu^2 > 0$ if $\mu^2 \rightarrow \infty$.*

As a consequence, neither AR, CLR or LM are valid tests under the combination of weak instruments and treatment effect heterogeneity. ■

In Figure 1, we plot the size of the AR, LM and CLR tests, used as (quasi-)tests for the TSLS hypothesis $\beta = \beta_0$, under the assumption of constant effects, and with treatment effect heterogeneity. In panel (a), we see that the AR, LM and CLR tests are valid with constant treatment effects at every level of instrument strength. In panel (b), we see that with treatment effect heterogeneity, all three tests overreject. We see that the degree of overrejection is increasing in instrument strength for AR and CLR and decreasing for LM.

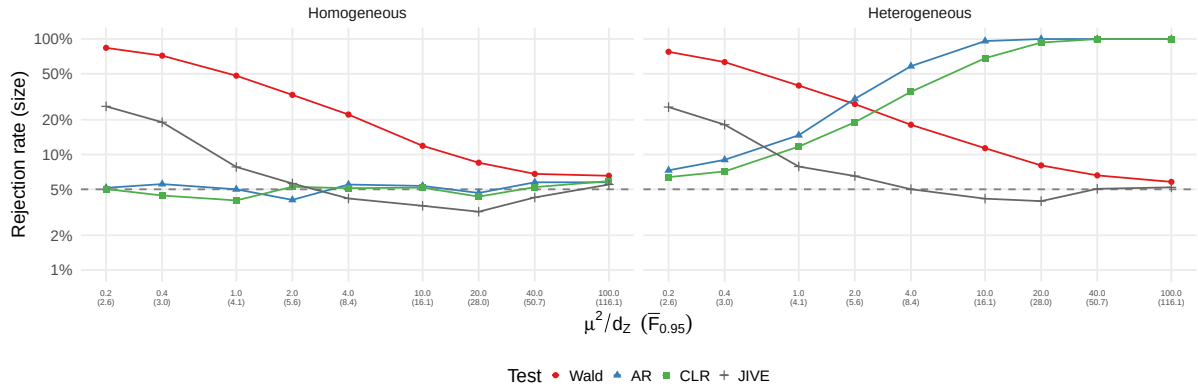


Figure 1: Overrejection of the Wald, AR and CLR Tests Under Heterogeneous Treatment Effects This figure plots the rate of over-rejection of the hypothesis $\beta = \beta_0$ when β is homogeneous, and when it is the TSLS/JIVE estimand (heterogeneous treatment effects), with weak instruments. The red line shows the Wald test, the blue line the [Anderson and Rubin \(1949\)](#) AR test and the green line the [Moreira \(2003\)](#) CLR test. The y axis shows the rejection rate (size), and the x axis enumerates instrument strength.

4. VALID INFERENCE ON THE TSLS ESTIMAND WITH WEAK INSTRUMENTS

We now turn to deriving a test that is valid under weak instruments with treatment effect heterogeneity.

The TLR Statistic The AR, LM and CLR tests are derived under the constant-effects hypothesis. We have derived a framework for imposing hypotheses on the TSLS estimand in terms of the asymptotic likelihood of $\hat{\tau}$. The resulting likelihood ratio statistic is a quadratic program with a quadratic constraint. In particular, it is a special case of the generalized trust region subproblem, and has a unique solution, as shown e.g. by [Stern and Wolkowicz \(1995\)](#). We record the result as a proposition.

Proposition TLR (The TSLS Likelihood Ratio Statistic). *The asymptotic likelihood ratio statistic for the hypothesis $\beta = \beta_0$ is given as*

$$\text{TLR}(\beta_0) = \min_{\tau_n \in \mathbb{R}^{2d}} n(\hat{\tau} - \tau_n)' \hat{\Sigma}^{-1} (\hat{\tau} - \tau_n) \quad s.t. \quad \tau_n' \hat{\Gamma}(\beta_0) \tau_n = 0$$

where $\hat{\Gamma}(\beta_0)$ is a consistent estimator for $\Gamma(\beta_0)$. The program has a unique solution,

$$\text{TLR}(\beta_0) = \sum_{j=1}^{2d_Z} \hat{W}_j^2 \left(\frac{\lambda^* \hat{\kappa}_j}{1 + \lambda^* \hat{\kappa}_j} \right)^2 \quad s.t. \quad \sum_{j=1}^{2d_Z} \frac{\hat{\kappa}_j \hat{W}_j^2}{(1 + \lambda^* \hat{\kappa}_j)^2} = 0$$

where $\hat{\kappa}_j$ denote the eigenvalues of $\hat{\Lambda}(\beta_0)$, $\hat{W}_j$ denote the elements of $\hat{W}$ and $\lambda^* \in [(\min_{j:\hat{\kappa}_j < 0} \hat{\kappa}_j)^{-1}, (\max_{j:\hat{\kappa}_j > 0} \hat{\kappa}_j)^{-1}]$ is unique.

Proof. See Appendix [A.5](#). ■

Limiting Distribution of TLR. Under maintained assumptions, the TLR statistic has a limiting distribution with two nuisance parameters. We record the result.

Proposition TLR-LIM (The TLR Limiting Distribution). *Under [Assumption IVX](#), the limiting distribution of the TLR statistic with weak instruments obtains as follows.*

$$\text{TLR}(\beta_0) \xrightarrow{d} \left(\sqrt{(1 - \varrho) Q_+} - \sqrt{\varrho Q_-} \right)^2 \quad (13)$$

where $Q_+ \sim \mathfrak{C}_{d_Z}^2(\varrho \xi)$, $Q_- \sim \mathfrak{C}_{d_Z}^2((1 - \varrho) \xi)$.

Proof. See Appendix [A.6](#). ■

It is useful to observe that the weak-instrument limiting distribution obtains the strong-instrument limiting distribution as $\xi \rightarrow \infty$. We record this, and some other useful boundary results as a proposition.

Proposition BC (Boundary Case Distributions).

1. As $\xi \rightarrow \infty$, we have $TLR(\beta_0) \xrightarrow{d} \mathfrak{C}_1^2$.
2. At $\xi = 0$, $\varrho \in \{0, 1\}$, we have $TLR(\beta_0) \xrightarrow{d} \mathfrak{C}_{d_Z}^2$.
3. At $\xi = 0$, $\varrho = 1/2$, we have $TLR(\beta_0) \xrightarrow{d} \left(\sqrt{\mathfrak{C}_{d_Z}^2} - \sqrt{\mathfrak{C}_{d_Z}^2} \right)^2 / 2$.

Proof. See Appendix A.7. ■

We now turn to using the above results to construct a uniformly valid test for the hypothesis $\beta = \beta_0$.

Conservative TLR Inference. The limiting distribution of the TLR statistic depends on two nuisance parameters: ϱ and ξ . We have a consistent estimator for ϱ , namely $\hat{\varrho} \equiv \sum_{k=1}^{d_Z} \hat{\kappa}_+^{(k)} / (\hat{\kappa}_+^{(k)} + |\hat{\kappa}_-^{(k)}|)$. We can use this estimator to compute worst-case quantiles. This gives a conservative test. We record the result as a proposition.

Proposition TLR-LFC (Least-favorable ξ Inference). *Let $c_{\varrho, 1-\alpha}(\xi)$ denote the $(1-\alpha)^{th}$ quantile of the distribution of the TLR estimator fixing (ϱ, ξ) . We then have:*

$$\mathbb{P} \left(TLR(\beta_0) \geq \max_{\xi \in \mathbb{R}^+} c_{\hat{\varrho}, 1-\alpha}(\xi) \right) \rightarrow \alpha$$

hence $TLR(\beta_0) \geq \max_{\xi \in \mathbb{R}^+} c_{\hat{\varrho}, 1-\alpha}(\xi)$ is a valid test for the hypothesis $\beta = \beta_0$.

Proof. See Appendix A.8. ■

As we show in Lemma STAT, $\hat{S}$ is a sufficient statistic for ξ . It follows that we can use $\hat{S}$ to perform inference and construct confidence intervals for the second nuisance parameter. This paves the way for a test in the spirit of Berger and Boos (1994) that retains power as the instrument grows stronger. We record the result.

Proposition TLR-BB (Berger-Boos Two-step Inference). *Let $\hat{\Xi}_{1-\alpha_1}$ denote a $1-\alpha_1$ confidence interval obtained by inverting $\hat{S}$, and let $\alpha_2 \equiv \alpha - \alpha_1$. Let $c_{\varrho, 1-\alpha_2}(\xi)$ denote the $(1-\alpha_2)^{th}$ quantile of the distribution of the TLR estimator. We then have:*

$$\mathbb{P} \left(TLR(\beta_0) \geq \max_{\xi \in \Xi_{1-\alpha_1}} c_{\hat{\varrho}, 1-\alpha_2}(\xi) \right) \rightarrow \alpha$$

hence $TLR(\beta_0) \geq \max_{\xi \in \Xi_{1-\alpha_1}} c_{\hat{\varrho}, 1-\alpha_2}(\xi)$ is a valid test for the hypothesis $\beta = \beta_0$.

Proof. See Appendix A.9. ■

The Berger-Boos Two-step test trades off a small price, α_2 , in order to guard against $\xi = 0$ when $\hat{S}$ is moderate and large. While the procedure turns out to perform well even with choices $\alpha_2 \approx 0$, it also means that the TLR test statistic will never obtain the strong-instrument limit, and always be less powerful than the TSLS Wald statistic in the face of weak instrument. In the following, we leverage results from [Elliott et al. \(2015\)](#) in order to construct a near power optimal test.

The TLR Test. Observe that the limiting distribution of the TLR statistic can be written on a conditional form. We record the result as a lemma.

Lemma COND (Conditional Distribution). *Let $\hat{R} = \hat{Q}_+/\hat{S}$ and $R \equiv Q_+/S$ where $S \sim \mathfrak{C}_{2d_Z}^2$. We then have*

$$\text{TLR}(\beta_0)/\hat{S} \mid \hat{S} \xrightarrow{d} \left(\sqrt{(1-\varrho)R} - \sqrt{\varrho R} \right)^2 \quad (14)$$

In Figure 2 we plot the size of the strong-instrument TLR test, the LFC TLR test and the Berger-Boos TLR test under worst-case endogeneity. We see that the strong-instrument test is invalid with weak instruments. We further see that the LFC TLR test is very conservative. The Berger-Boos corrected TLR test is valid and close to exact.

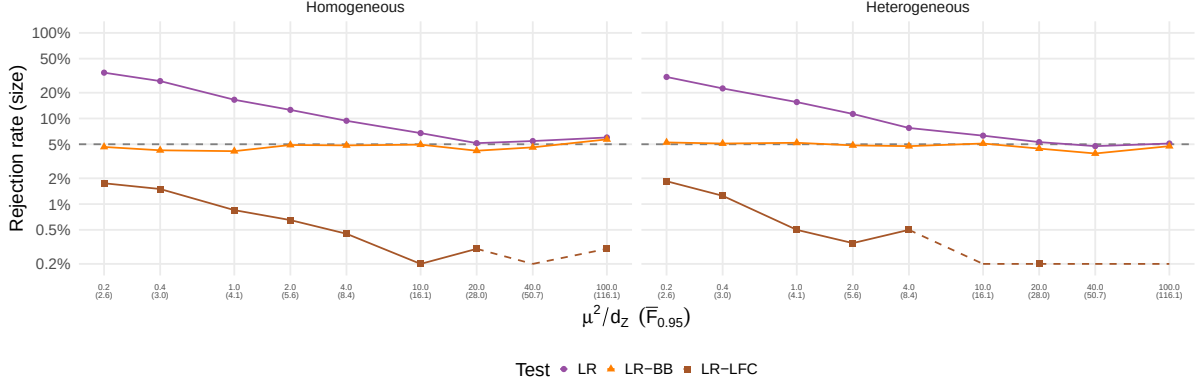


Figure 2: Size of the Berger-Boos corrected LR Test. This figure plots the size of the Berger-Boos corrected LR test for the hypothesis $\beta = \beta_0$ when β is homogeneous, and when it is the TSLS/JIVE estimand (heterogeneous treatment effects), with weak instruments. The orange line shows the corrected LR test, the purple line shows the uncorrected LR test, and the brown line shows the LFC-LR test. The y axis shows the rejection rate (size), and the x axis enumerates instrument strength.

In Figure 3 we plot the power of the Berger-Boos TLR test under worst-case endogeneity. We compare it to the power of the TSLS and Jacckknife IV Wald test, to obtain a notion of the degree to which the power of our test is on the level, and it is.

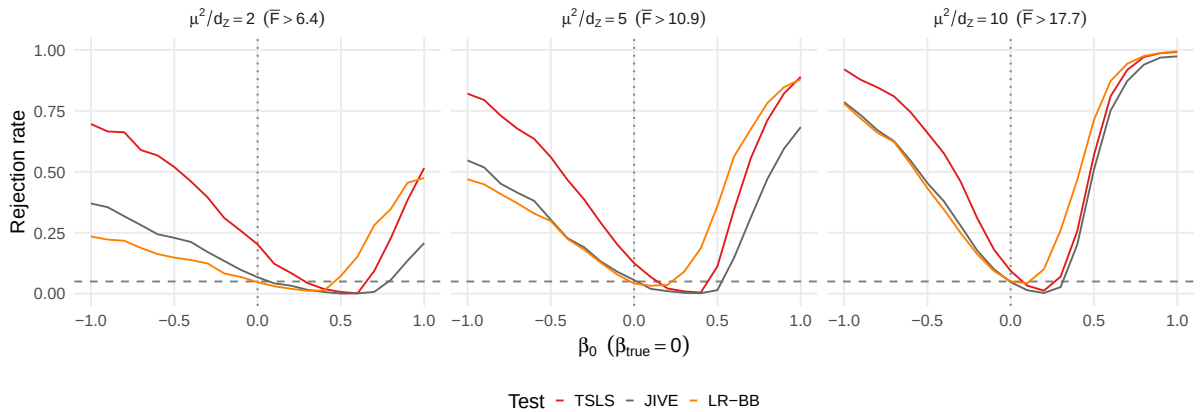


Figure 3: Power of the Berger-Boos corrected LR Test. This figure plots the power against alternative of the Berger-Boos corrected LR test for the hypothesis $\beta = \beta_0$. The orange line shows the corrected LR test, the red line shows the TSLS Wald test and the gray line the JIVE Wald test. The y axis shows the rejection rate, and the x axis enumerates distance from the true parameter.

5. EXTENSIONS

General Limiting Distribution with Weak Instruments and Weak Exogeneity.

Assume now we instead imposed a weaker assumption than [Assumption IVX](#). In particular, consider the following assumption.

Assumption IVW (Weak Exogeneity). $\mathbb{E}[Z_i U_i] = 0_{d_Z}$.

Under the weaker assumption, [Lemma EH](#) does not obtain. It thus follows that we obtain a less tractable version of [Proposition TLR-LIM](#).

Proposition WI (Weak-instrument Limiting Distribution). *Under [Assumption IVW](#), the limiting distribution of the TLR statistic with weak instruments is given by:*

$$\text{TLR}(\beta_0) \xrightarrow{d} \sum_{j=1}^{2d_Z} W_j^2 \left(\frac{\lambda^* \kappa_j}{1 + \lambda^* \kappa_j} \right)^2 \quad s.t. \quad \sum_{j=1}^{2d_Z} \kappa_j w_j^2 = 0 \quad (15)$$

where $W \sim \mathfrak{N}(w, \mathbf{I}_{2d_Z})$ and W_j are the components of this vector.

Proof. See [Appendix A.11](#). ■

Without any further structure, this problem has $4d_Z - 1$ nuisance parameters. We have consistent estimators for all κ_j . However, for w we only have the $2d_Z$ -dimensional statistic $\hat{W}$. It is possible to perform a version of the TLR-BB test.

5.0.0.1. Notes:

- I mu2, nu2 osv resultat: Skriv in GMM estimator istedenfor TSLS (nei, men husk på det evt til extension). hetskd versjon

6. CONCLUSION

1. Short summary

Bibliography

- Anderson, T. W. and Rubin, H. (1949). Estimation of the Parameters of a Single Equation in a Complete System of Stochastic Equations. *The Annals of Mathematical Statistics*, 20(1):46–63.
- Andrews, D. W. K., Moreira, M. J., and Stock, J. H. (2006). Optimal Two-Sided Invariant Similar Tests for Instrumental Variables Regression. *Econometrica*, 74(3):715–752.
- Andrews, I. (2016). Conditional Linear Combination Tests for Weakly Identified Models. *Econometrica*, 84(6):2155–2182.
- Angrist, J. D., Graddy, K., and Imbens, G. W. (2000). The Interpretation of Instrumental Variables Estimators in Simultaneous Equations Models with an Application to the Demand for Fish. *The Review of Economic Studies*, 67(3):499–527.
- Angrist, J. D. and Imbens, G. W. (1995). Two-Stage Least Squares Estimation of Average Causal Effects in Models with Variable Treatment Intensity. *Journal of the American Statistical Association*, 90(430):431–442.
- Baiocchi, M., Cheng, J., and Small, D. S. (2014). Tutorial in Biostatistics: Instrumental Variable Methods for Causal Inference*. *Statistics in medicine*, 33(13):2297–2340.
- Berger, R. L. and Boos, D. D. (1994). P Values Maximized Over a Confidence Set for the Nuisance Parameter. *Journal of the American Statistical Association*, 89(427):1012–1016.
- Bound, J., Jaeger, D. A., and Baker, R. M. (1995). Problems with Instrumental Variables Estimation When the Correlation Between the Instruments and the Endogenous Explanatory Variable is Weak. *Journal of the American Statistical Association*, 90(430):443–450.
- Chao, J. C. and Swanson, N. R. (2005). Consistent Estimation with a Large Number of Weak Instruments. *Econometrica*, 73(5):1673–1692.
- Chao, J. C., Swanson, N. R., Hausman, J. A., Newey, W. K., and Woutersen, T. (2012). Asymptotic Distribution of Jive in a Heteroskedastic Iv Regression with Many Instruments. *Econometric Theory*, 28(1):42–86.
- Elliott, G., Müller, U. K., and Watson, M. W. (2015). Nearly Optimal Tests When a Nuisance Parameter Is Present Under the Null Hypothesis. *Econometrica*, 83(2):771–811.

- Evdokimov, K. S. and Kolesár, M. (2018). Inference in Instrumental Variables Analysis with Heterogeneous Treatment Effects. Working Paper.
- Finlay, K. and Magnusson, L. M. (2009). Implementing Weak-Instrument Robust Tests for a General Class of Instrumental-Variables Models. *The Stata Journal*, 9(3):398–421.
- Hansen, L. P. (1982). Large Sample Properties of Generalized Method of Moments Estimators. *Econometrica*, 50(4):1029–1054.
- Kleibergen, F. (2007). Generalizing weak instrument robust IV statistics towards multiple parameters, unrestricted covariance matrices and identification statistics. *Journal of Econometrics*, 139(1):181–216.
- Kolesár, M. (2013). Estimation in an Instrumental Variables Model With Treatment Effect Heterogeneity. Working Paper.
- Mikusheva, A. and Sun, L. (2022). Inference with Many Weak Instruments. *The Review of Economic Studies*, 89(5):2663–2686.
- Mikusheva, A. and Sun, L. (2024). Weak Identification with Many Instruments. arXiv Working Paper 2308.09535.
- Mogstad, M., Torgovitsky, A., and Walters, C. R. (2021). The Causal Interpretation of Two-Stage Least Squares with Multiple Instrumental Variables. *American Economic Review*, 111(11):3663–3698.
- Moreira, M. J. (2003). A Conditional Likelihood Ratio Test for Structural Models. *Econometrica*, 71(4):1027–1048.
- Moreira, M. J. (2009). Tests with correct size when instruments can be arbitrarily weak. *Journal of Econometrics*, 152(2):131–140.
- Sargan, J. D. (1958). The Estimation of Economic Relationships using Instrumental Variables. *Econometrica*, 26(3):393–415.
- Staiger, D. and Stock, J. H. (1997). Instrumental Variables Regression with Weak Instruments. *Econometrica*, 65(3):557–586.
- Stern, R. J. and Wolkowicz, H. (1995). Indefinite Trust Region Subproblems and Non-symmetric Eigenvalue Perturbations. *SIAM Journal on Optimization*, 5(2):286–313.

- Stock, J. H., Wright, J. H., and Yogo, M. (2002). A Survey of Weak Instruments and Weak Identification in Generalized Method of Moments. *Journal of Business & Economic Statistics*, 20(4):518–529.
- Stock, J. H. and Yogo, M. (2005). Testing for Weak Instruments in Linear IV Regression. In Stock, J. H. and Andrews, D. W. K., editors, *Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg*, pages 80–108. New York.
- Yap, L. (2025). Inference with Many Weak Instruments and Heterogeneity. arXiv Working Paper 2408.11193.

A. PROOFS

A.1. Proof of Lemma EH

Proof. TBD. See proof from C. Notes for signature part:

The matrix $\Gamma(\beta_0)$ has signature (d_Z, d_Z) , i.e. d_Z positive and d_Z negative eigenvalues. $\Lambda(\beta_0)$ is congruent to $\Gamma(\beta_0)$, hence has the same signature. Use exact expression for eigenvalues of Gamma beta0.³ ■

A.2. Proof of Lemma STAT

Proof. TBD. Trivial. ■

A.3. Proof of Lemma LEG-CTE

Proof. TBD. ■

A.4. Proof of Lemma LEG-HTE

Proof. TBD. ■

A.5. Proof of Proposition TLR

Proof. TBD. See slides. ■

A.6. Proof of Proposition TLR-LIM

Proof. TBD. See: Written before the redefinition of ξ (used to be ξ_+ only).

we can write the secular equation along the weak instruments limiting sequence as:

$$\frac{\kappa_+^*}{(1 + \lambda^* \kappa_+^*)^2} \sum_{\kappa_j > 0} W_j^2 - \frac{|\kappa_-^*|}{(1 - \lambda^* |\kappa_-^*|)^2} \sum_{\kappa_j < 0} W_j^2 = 0$$

Observe now that

$$Q_+ \equiv \sum_{\kappa_j > 0} W_j^2 \sim \mathfrak{C}_{d_Z}^2(\xi_+) \quad \text{and} \quad Q_- \equiv \sum_{\kappa_j < 0} W_j^2 \sim \mathfrak{C}_{d_Z}^2(\xi_-) \quad (16)$$

³Observe that since Σ is defined by $\Sigma_{\hat{\delta}} \equiv \Sigma_{ZZ}^{-1} \mathbb{E}[\tilde{Z}_i \tilde{Z}_i' W_i^2] \Sigma_{ZZ}^{-1}$, $\Sigma_{\hat{\gamma}} \equiv \Sigma_{ZZ}^{-1} \mathbb{E}[\tilde{Z}_i \tilde{Z}_i' V_i^2] \Sigma_{ZZ}^{-1}$, and $\Sigma_{\hat{\delta}, \hat{\gamma}} \equiv \Sigma_{ZZ}^{-1} \mathbb{E}[\tilde{Z}_i \tilde{Z}_i' W_i V_i] \Sigma_{ZZ}^{-1}$, we can write $\Lambda(\beta_0)$ as

$$\Lambda(\beta_0) = \mathbb{E} \left[\left[\begin{array}{cc} W_i^2 & W_i V_i \\ W_i V_i & V_i^2 \end{array} \right] \otimes \tilde{Z}_i \tilde{Z}_i' \right]^{\frac{1}{2}} \left[\begin{array}{cc} 0 & 1 \\ 1 & -2\beta_0 \end{array} \right] \otimes \iota_{d_Z} \mathbb{E} \left[\left[\begin{array}{cc} W_i^2 & W_i V_i \\ W_i V_i & V_i^2 \end{array} \right] \otimes \tilde{Z}_i \tilde{Z}_i' \right]^{\frac{1}{2}}$$

where $\tilde{Z}_i \equiv \Sigma_{ZZ}^{-\frac{1}{2}} Z_i$ is the whitened instrument. The signature follows since $\begin{bmatrix} 0 & 1 \\ 1 & -2\beta_0 \end{bmatrix}$ has d_Z identical pairs of eigenvalues taking the values $-\beta_0 + \sqrt{\beta_0^2 - 1}$ and $\Gamma(\beta_0)$ and $\Lambda(\beta_0)$ are congruent to this matrix.

where Q_+, Q_- are independent, and under the null we have along the asymptotic sequence, where $\sqrt{n}w_n = w$, that:

$$\sum_{j=1}^{2d_Z} \kappa_j w_j^2 = 0 \iff |\kappa_-| \sum_{\kappa_j < 0} w_j^2 = \kappa_+ \sum_{\kappa_j > 0} w_j^2$$

and by the properties of non-central chi-squared distributions, this implies $\varrho^* \xi_- = \xi_+ \equiv \xi$. The secular equation thus simplifies as:

$$\frac{Q_+}{(1 + \lambda^* \kappa_+^*)^2} = \frac{\varrho^* Q_-}{(1 - \lambda^* |\kappa_-^*|)^2}$$

Taking the root and solving for $\lambda^* \kappa_+^*$, we obtain:

$$\lambda^* \kappa_+^* = \frac{\sqrt{Q_+} - \sqrt{\varrho^* Q_-}}{\varrho^* \sqrt{Q_+} + \sqrt{\varrho^* Q_-}}$$

Under ?? the LR statistic obtains:

$$\text{TLR}(\beta_0) \xrightarrow{d} \left(\frac{\lambda^* \kappa_+^*}{1 + \lambda^* \kappa_+^*} \right)^2 Q_+ + \left(\frac{\lambda^* \kappa_-^*}{1 + \lambda^* \kappa_-^*} \right)^2 Q_-$$

Inserting for [equation \(17\)](#) in this limit, we obtain:

$$\text{TLR}(\beta_0) \xrightarrow{d} \left(\frac{\sqrt{Q_+} - \sqrt{\varrho^* Q_-}}{(1 + \varrho^*) \sqrt{Q_+}} \right)^2 Q_+ + \left(\frac{\varrho^* (\sqrt{Q_+} - \sqrt{\varrho^* Q_-})}{(1 + \varrho^*) \sqrt{\varrho^* Q_-}} \right)^2 Q_-$$

This simplifies to:

$$\text{TLR}(\beta_0) \xrightarrow{d} \frac{(\sqrt{Q_+} - \sqrt{\varrho^* Q_-})^2}{1 + \varrho^*} \tag{17}$$

where $Q_+ \sim \mathfrak{C}_{d_Z}^2(\xi)$ and $Q_- \sim \mathfrak{C}_{d_Z}^2(\xi/\varrho^*)$. ■

A.7. Proof of *Proposition BC*

Proof. TBD. See pdf note for (a)–(d). ■

A.8. Proof of *Proposition TLR-LFC*

Proof. TBD. Trivial. ■

A.9. Proof of *Proposition TLR-BB*

Proof. TBD. Semi-trivial. ■

A.10. Proof of Lemma COND

Proof. TBD. ■

A.11. Proof of Proposition WI

Proof. TBD. Without Assumption IVX the eigenvalue homogeneity result Lemma EH fails, so the secular equation does not collapse to the two-eigenvalue form. Argument follows along the lines of the proof of Proposition TLR-LIM but without the simplification step. ■

Bibliography